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## Introduction

This guide explores how selected Artificial Intelligence (AI) tools can support learners with neurodevelopmental disorders, with a focus on the study of Modern Languages. As digital technologies become increasingly integrated into education, AI offers new opportunities to reduce barriers to learning, promote independence, and provide personalised support tailored to individual needs. For students with conditions such as dyslexia, autism spectrum disorder (ASD), attention deficit hyperactivity disorder (ADHD), and other neurodevelopmental differences, AI tools can help make language learning more accessible and engaging.

Modern Language assessments often require students to demonstrate a broad range of skills, including reading, writing, listening, speaking, grammar, vocabulary acquisition, and cultural understanding. These tasks can present additional challenges for neurodivergent learners, particularly where difficulties with processing, organisation, memory, concentration, or communication are involved. AI technologies can provide practical assistance by offering real-time feedback, personalised learning pathways, speech-to-text and text-to-speech capabilities, translation support, vocabulary reinforcement, and study organisation tools.

The intended audience for this guide is undergraduate students of Cardiff University, with a focus on those studying Modern Languages. This guide covers a wide range of functionality provided by selected AI tools, how to use them, and how they can help Modern Languages students with task preparation. This guide is a baseline for how AI tools can be used, and in no way covers every single use of every single tool covered, just ways in which we have identified could help students. This guide provides a baseline overview of how AI tools may be used to support neurodivergent students. It does not seek to cover every possible application of the tools included, but rather highlights selected uses that were identified as potentially beneficial across different aspects of the student experience. The development of the guide was further informed by feedback from volunteer students within the Modern Languages (MLANG) community, whose reflections and lived experiences helped refine its content, relevance, and accessibility.

AI tools should be viewed as supplementary support mechanisms rather than replacements for existing reasonable adjustments, specialist support services, or inclusive teaching practices. The effectiveness of any tool will vary according to individual needs, preferences, and contexts, and students should be encouraged to identify the approaches that best support their learning.

## Why these disorders?

### What is neurodiversity

Neurodiversity is the concept of that there is natural variation in the ways that the human brain works, signifying that there is no one “correct” way to think, learn, or process information.

*“Neurodiversity is the concept that there is natural variation in how people’s brains work, with no single “correct” way, and that a wide range of perceptions and responses to the world should be accepted and encouraged” – (Child Mind Institute, 2026).*

### What disorders are we covering, and why?

#### Dyslexia

##### **Evidence**

Dyslexia effects reading accuracy, spelling, phonological processing and vocabulary. These difficulties can impact reading comprehension, grammar, and written language production in Modern Languages. (University of Strathclyde, n.d.)

##### **Rationale for inclusion**

Modern Language relies heavily on reading, writing, spelling, and pronunciation. AI tools that can perform text-to-speech, speech recognition, and grammar support that can provide huge benefits for those who struggle.

##### **Possible link to assessments/ tasks**

- Writing exams
- Reading exams
- Vocabulary
- Pronunciation activities

## Dyspraxia

### ***Evidence***

Dyspraxia can affect motor coordination, planning and organisation, handwriting speed and notetaking. People with dyspraxia may also struggle with planning (e.g. creating a plan for tasks and projects) (Dyspraxia Foundation (2012))

### ***Rationale for inclusion***

Written tasks and language learning activities often require extensive notetaking and written production, AI based speech-to-text may reduce these barriers. AI tools can also be used to effectively plan tasks and create timelines.

### ***Possible link to assessment/tasks***

- Notetaking
- Written Tasks
- Project Planning

## Attention Deficit Hyperactivity Disorder (ADHD)

### ***Evidence***

ADHD can affect concentration, working memory, time management and task completion. Learners may struggle to remember vocabulary, organise revision, or sustain attention span during lengthy lessons and tasks. (Royal Borough of Windsor and Maidenhead, n.d.)

### ***Rationale for inclusion***

Modern Languages tasks require sustained focus and structured responses. AI tools can support planning, revision planning, resource organisation, and personalised learning.

### ***Possible links to assessments/ tasks***

- Writing task
- Listening tasks
- Revision and coursework management

## Autism (autism spectrum disorder)

### ***Evidence***

Autistic learners may experience challenges with social communication, interpreting figurative language, or understanding contextual clues. Language learning often involves cultural interpretation and communication skills. (Reachout ASC (n.d.)).

### ***Rationale for inclusion***

Modern Languages requires both linguistic and social communication competencies. AI tools can provide structured explanations, conversational practise and personalised learning environments.

### ***Possible links to assessments/ tasks***

- Speaking assessments
- Cultural interpretation tasks
- Group tasks

## Processing disorders

### ***Evidence***

Processing disorders can affect auditory, visual, and speed of processing. This means that tasks such as reading, listening, following instructions, vocabulary retention, and language comprehension may be challenging, particularly in the study of Modern Languages. (National Center for Learning Disabilities (n.d.))

### ***Rationale for inclusion***

Modern Language Learning requires learners to quickly process spoken language and written information, remember new vocabulary, and interpret meaning from context. AI tools that provide text-to-speech, content summarisation, and personalised pacing can help reduce barriers for learners with processing difficulties

### ***Possible link to assessments/ tasks***

- Writing exams
- Reading exams
- Vocabulary
- Multi step instructions
- Note taking

## Why is a recourse like this relevant to Cardiff University, and education as a whole?

Inclusive education and recognition of learners at Cardiff University with neurodevelopmental disorders is very important. Neurodivergence is real and impacts people in all different areas of life. Cardiff University has 7% (1,980) of students that have disclosed that they are neurodivergent as of the 25/26 academic year, along with 7.8% (40) of MLANG students that have also disclosed the fact they are neurodivergent.

With this number of students that have disclosed that they are neurodivergent, it is still very common for people to get to the point of being in the first year of their university course, unaware that they are neurodivergent, and therefore do not get the support they need. Promoting this topic is important, as it will allow for people to get support they did not know they needed, helping further contribute to equality in education.

Statistics from higher education state that neurodiversity is increasingly important in the sector, with recent statistics showing that 22% of UK students have a neurodivergent diagnosis, with up to 28% identifying as neurodivergent in some way. This contributes to an overrepresentation of neurodivergence among students compared to the general population, of which only 15-20% are believed to be neurodivergent (Eves, 2026). This highlights the growing importance of neurodiversity in higher education and reinforces the need for targeted strategies and resources that enable neurodivergent students to thrive academically

## Why these tools?

We conducted a comprehensive review of a wide range of tools to identify those most suitable for neurodivergent students at Cardiff University. The evaluation began with an initial shortlist of 11 tools, which were assessed against a set of rigorous criteria to determine their suitability and effectiveness. The criteria used in the evaluation are outlined below:

### Evaluation criteria and rating of selected AI tools

What we are evaluating these tools for:

| Criterion                     | Definition                                                                                                | Indicators                                                                                                | Rating (1-5)                |
|-------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------|
| Accessibility                 | The extent to which this tool is accessible for the targeted students with neurodevelopmental disorders   | Text-to-speech, clear language, keyboard accessibility, simple interface, adjustable text size, captions. | 1 = Poor<br>5 = Outstanding |
| Personalisation               | The features provided which allow the users to change the app, to promote a more user specific experience | Tailored explanations, adjustable difficulty, personalised feedback, response customisation.              | 1 = Poor<br>5 = Outstanding |
| Cognitive load and management | The extent that the tool can help reduce unnecessary mental effort                                        | Step-by-step guidance, concise responses, chunking of information, minimal distractions.                  | 1 = Poor<br>5 = Outstanding |
| Pedagogical Usefulness        | How well the AI tool supports meaningful learning and problem solving.                                    | Accurate explanations, scaffolding, constructive feedback, encourages critical thinking.                  | 1 = Poor<br>5 = Outstanding |

The criteria for scoring:

| Score | Meaning                                                              |
|-------|----------------------------------------------------------------------|
| 1     | Does not meet the criteria, not fit or purpose                       |
| 2     | Meets the criteria to a limited extent, improvements would be needed |
| 3     | Meets the criteria moderately, basic use possible                    |
| 4     | Meets the criteria, intended use very possible                       |
| 5     | Meets the criteria fully, intended use is excellently provided.      |

First, we started creating a series of relevant categories, aligned to features in an app that would be useful to our target audience. We then created a criterion for scoring across characteristics of every category we have defined.

*Below are the scoring criteria:*

Accessibility:

| Indicator                                 | What to look for                                                     | Rating 0 – 5   |
|-------------------------------------------|----------------------------------------------------------------------|----------------|
| Clear and readable interface              | Clean layout, readable fonts, minimal clutter                        |                |
| Accessible communication                  | Uses plain language and can simplify explanations                    |                |
| Multiple ways to interact                 | Supports text, voice, images, or other input/output methods          |                |
| Computability with accessibility features | Works with screen readers, keyboard navigation, text-to-speech, etc. |                |
| Sensory accessibility                     | Avoids overwhelming visuals or unnecessary distractions              |                |
| Total                                     |                                                                      | Points average |

Personalisation:

| Indicator                         | What to look for                                                       | Points 0 – 5   |
|-----------------------------------|------------------------------------------------------------------------|----------------|
| Adaptive explanations             | can simplify, expand, or rephrase content                              |                |
| Adjustable response style         | User can request different formats, lengths, or levels of detail       |                |
| Personalised feedback             | Feedback reflects the learner's responses rather than generic advice   |                |
| Supports individual learning pace | Learners can revisit, repeat, or build on previous explanations        |                |
| Flexibility                       | Can generate examples, quizzes, summaries, or alternative explanations |                |
| Total                             |                                                                        | Points average |

Cognitive load management

| Indicator             | What to look for                                              | Points 0 – 5 |
|-----------------------|---------------------------------------------------------------|--------------|
| Chunking information  | Breaks information into manageable sections                   |              |
| Step by step guidance | Explains tasks sequentially                                   |              |
| Concise explanations  | Avoids unnecessary complexity or information overload         |              |
| Organisation          | Uses headings, bullet points, summaries, or logical structure |              |
| Attention support     | Keeps responses focused and avoids irrelevant information     |              |
| Total                 |                                                               | Points       |

Pedagogical usefulness

| Indicator                     | What to look for                                                      | Points 0 – 5 |
|-------------------------------|-----------------------------------------------------------------------|--------------|
| Concept explanation           | Explains reasoning rather than only giving answers                    |              |
| Constructive feedback         | Identifies mistakes and explains how to improve                       |              |
| Encourages critical thinking  | Prompts learners to think rather than simply copy                     |              |
| Supports independent learning | Helps learners solve problems themselves                              |              |
| Learning reinforcement        | Provides examples, practice questions, summaries, or revision support |              |
| Total                         |                                                                       | Points       |

From this point, we made note of an average score from a section, and add it to every other average score to get a total score for the app. From this we were able to see how every tool we analysed compared with each other, and which tools aligned best with the needs of students with neurodevelopmental disorders.

*Below is the selection of tools we analysed, along with the task they were best aligned for:*

| Reading                                                                                                                                                                       | Writing                                                                                              | Listening                                                                        | Speaking                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Perplexity</li> <li>• Notebook LM</li> <li>• Knowt</li> <li>• Language tool</li> <li>• Wolfram alpha</li> <li>• Photomath</li> </ul> | <ul style="list-style-type: none"> <li>• Microsoft Copilot (w/ voice)</li> <li>• Quillbot</li> </ul> | <ul style="list-style-type: none"> <li>• Google Gemini (w/ live mode)</li> </ul> | <ul style="list-style-type: none"> <li>• Google Gemini (w/ live mode)</li> </ul> |

*Below is the result from our analysis:*

*Final scores:*

| AI tool                      | Score     |
|------------------------------|-----------|
| Perplexity                   | 16.4 / 20 |
| Notebook LM                  | 15.6 / 20 |
| Knowt                        | 15.6 / 20 |
| Quillbot                     | 16.5 / 20 |
| Language tool                | 12.9 / 20 |
| Wolfram alpha                | 10.8 / 20 |
| Photo math                   | 11.8 / 20 |
| Microsoft Copilot (w/ voice) | 16.2 / 20 |
| Google Gemini (w/ live mode) | 16.5 / 20 |

*Score breakdown:*

| Score range | Description                                                    |
|-------------|----------------------------------------------------------------|
| 0 – 8.9     | Not fit for purpose                                            |
| 9 – 11.9    | Fit for purpose but may have limited use case.                 |
| 12 – 13.9   | Fit for purpose                                                |
| 14 – 15.9   | Fit for purpose, has great features, and a good use case       |
| 16 or more  | Fit for purpose, has excellent features, and a great use case. |

From this we were able to create a further shortlist of tools:

Cardiff University Provide Gemini Notebook, Google Gemini, and Microsoft Copilot. This also helped us when deciding which tools to shortlist.

*The final shortlisted tools:*

| Reading                                                                                                        | Writing                                                                                              | Listening and speaking                                                           |
|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Perplexity</li> <li>• Notebook LM</li> <li>• Language tool</li> </ul> | <ul style="list-style-type: none"> <li>• Microsoft Copilot (w/ voice)</li> <li>• Quillbot</li> </ul> | <ul style="list-style-type: none"> <li>• Google Gemini (w/ live mode)</li> </ul> |

## How did LanguageTool perform in our analysis?

### Language Tool

#### Accessibility:

| Indicator                                 | What to look for                                                     | Rating 0 – 5        |
|-------------------------------------------|----------------------------------------------------------------------|---------------------|
| Clear and readable interface              | Clean layout, readable fonts, minimal clutter                        | 3                   |
| Accessible communication                  | Uses plain language and can simplify explanations                    | 4                   |
| Multiple ways to interact                 | supports text, voice, images, or other input/output methods          | 1                   |
| Computability with accessibility features | Works with screen readers, keyboard navigation, text-to-speech, etc. | 2                   |
| Sensory accessibility                     | Avoids overwhelming visuals or unnecessary distractions              | 4                   |
| Total                                     |                                                                      | Points average: 2.8 |

#### Personalisation:

| Indicator                         | What to look for                                                       | Points 0 – 5               |
|-----------------------------------|------------------------------------------------------------------------|----------------------------|
| Adaptive explanations             | can simplify, expand, or rephrase content                              | 1                          |
| Adjustable response style         | User can request different formats, lengths, or levels of detail       | 2                          |
| Personalised feedback             | Feedback reflects the learner's responses rather than generic advice   | 4                          |
| Supports individual learning pace | Learners can revisit, repeat, or build on previous explanations        | 4                          |
| Flexibility                       | Can generate examples, quizzes, summaries, or alternative explanations | N/A                        |
| Total                             |                                                                        | Points average: 2.75 (2.8) |

#### Cognitive load management:

| Indicator             | What to look for                                              | Points 0 – 5      |
|-----------------------|---------------------------------------------------------------|-------------------|
| Chunking information  | Breaks information into manageable sections                   | N/A               |
| Step by step guidance | Explains tasks sequentially                                   | N/A               |
| Concise explanations  | Avoids unnecessary complexity or information overload         | 4                 |
| Organisation          | Uses headings, bullet points, summaries, or logical structure | 4                 |
| Attention support     | Keeps responses focused and avoids irrelevant information     | 4                 |
| Total                 |                                                               | Points average: 4 |

#### Pedagogical usefulness:

| Indicator                     | What to look for                                                      | Points 0 – 5        |
|-------------------------------|-----------------------------------------------------------------------|---------------------|
| Concept explanation           | Explains reasoning rather than only giving answers                    | N/A                 |
| Constructive feedback         | Identifies mistakes and explains how to improve                       | 3                   |
| Encourages critical thinking  | Prompts learners to think rather than simply copy                     | 3                   |
| Supports independent learning | Helps learners solve problems themselves                              | 4                   |
| Learning reinforcement        | Provides examples, practice questions, summaries, or revision support | N/A                 |
| Total                         |                                                                       | Points average: 3.3 |

SCORE : 12.9 / 20

## Disclaimers

- Always use your Cardiff University email address to access any AI tools recommended by the University. Where available, use the University-provided or University-approved versions of these tools. This helps ensure that your data is protected under Cardiff University's security, privacy and data protection arrangements.
- Avoid sharing personal, confidential, or sensitive information with AI tools whenever possible. This includes information about yourself, other students, staff members or any data that could identify an individual.
- The advice provided in this guide, and in any related University guidance, is intended to support your learning and should be considered guidance rather than a set of mandatory rules. You should always use your own judgement and follow any instructions provided by your module leaders or the University.
- AI-generated content is not always accurate. AI tools can sometimes produce incorrect, misleading, incomplete, or fabricated information (often referred to as "hallucinations"). You should carefully review, verify and critically evaluate all AI-generated outputs before using them in your studies.
- Use AI as a learning aid, not as a substitute for your own work. AI can help you understand concepts, organise ideas, practise language skills and receive feedback, but it should not be used to plagiarise or misrepresent work as your own. Always follow Cardiff University's policies on academic integrity and appropriate use of AI.

# LanguageTool

## What disorders can this tool help with?

### ADHD

This guide is informed by a comparison of LanguageTool capabilities and the challenges that learners with ADHD may experience in academic settings. The sections below explain how the LanguageTool tasks included throughout this guide can support learners with ADHD in managing their studies more effectively.

Based on the finding of this research, tasks that can be completed with LanguageTool, and its characteristics to issues students with ADHD may experience. Below will describe how the following tasks we have set out further along with this guide, involving LanguageTool, can help learners with ADHD.

### Drafting texts to catch grammar and spelling errors

Tasks may be harder to complete or start for those who have ADHD. This tool allows for the task of proofreading to be assisted, along with automated explanations behind why parts of a text may change, and what parts of a text may mean.

### Dyslexia

This guide is informed by a comparison of LanguageTool capabilities and the challenges that learners with dyslexia may experience in academic settings. The sections below explain how the LanguageTool tasks included throughout this guide can support learners with dyslexia in managing their studies more effectively.

Based on the findings of this research, we have compared tasks that can be completed with Language Tool, and its characteristics to issues students with dyslexia may experience. Below will describe how the following tasks we have set out further along with this guide, involving LanguageTool, can help learners with dyslexia.

### Drafting texts to catch grammar and spelling errors

It is common for learners with dyslexia to have complications tasks such as spelling and proofreading. This tool may be able to help students with dyslexia, by acting as a spelling and grammar assistant with writing, even in foreign languages.

## Modern languages assessments/ tasks this tool can help with

### *Essays and writing practise*

This tool can help with essays and writing practise in foreign languages. LanguageTool can do this by being able to catch spelling and grammar errors in an array of different languages.

## Desktop

### Setting up/ signing in

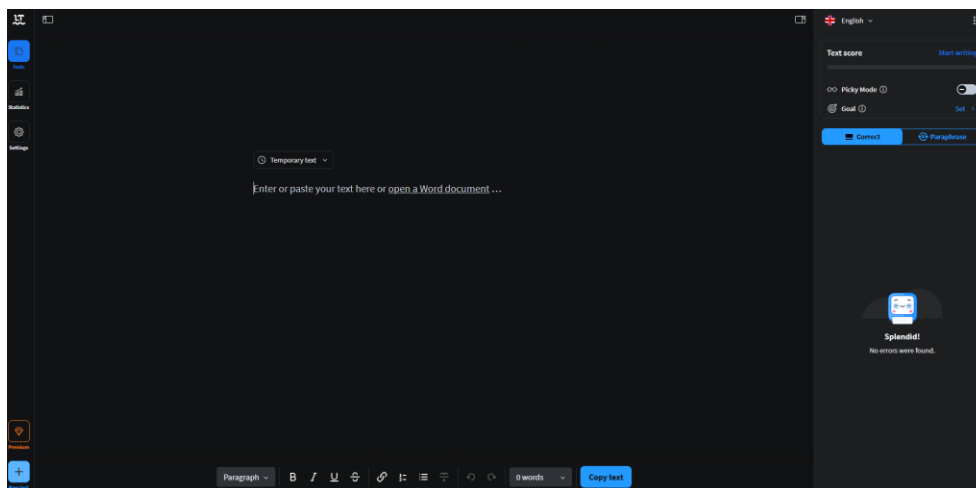
#### *How to do this*

You can access language Tool but simply searching “LanguageTool” into Google Chrome and select the first appropriate result. From that point you can sign in with an email of your choice. While Cardiff University does not currently provide access to this service, its potential to support neurodivergent learners is recognised, and it is therefore considered as a useful resource.

*Disclaimer: Where possible, users are encouraged to sign in to or create accounts for the recommended AI tools using their Cardiff University email address.*

#### *After set up/ signing in*

*You should be greeted with the following interface:*

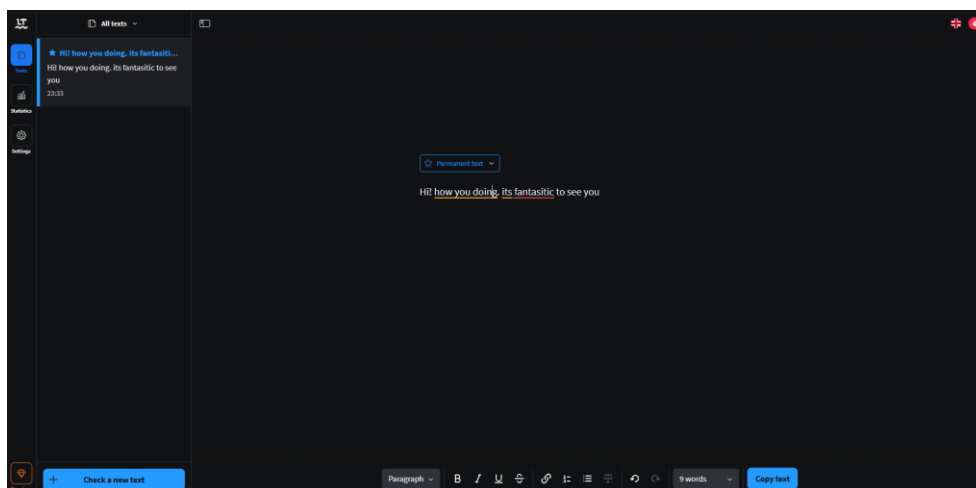


From this point, you have access to LanguageTool. You can type or paste your text into the text box that takes up most of the interface, or more specifically in front of “Enter or paste your text here or open a Word document”, the “Enter Word document” part is a hyperlink that will direct you to file explorer, where you can upload a Word document.

## Drafting texts to catch grammar and spelling errors

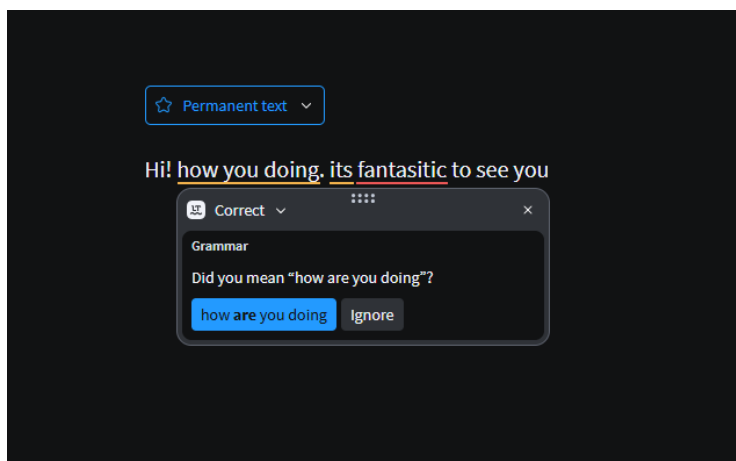
In Language tool, you can draft text by directly writing into LanguageTool. You can select if you want this text to be saved, by the drop-down menu, that allows you to switch between “Temporary text” and “Permanent text”. Making a text permanent will allow it to be saved. All permanent texts can be access by clicking the “Texts” Button in the left menu. While you do this you will receive pointers from LanguageTool about your spelling and grammar.

*For reference:*



Here you can see that the text is being saved in the left menu but is also receiving grammar and spelling suggestions.

*For reference:*



Language tool allows this to be used for a whole host of other languages, not just English.

## References

*Child Mind Institute (2026) What is neurodiversity? Available at:*

*<https://childmind.org/article/what-is-neurodiversity/> (Accessed: 3 September 2026).*

*Eves, L. (2026) Neuroinclusive teaching in higher education. Higher Education Policy Institute. Available at: <https://www.hepi.ac.uk/2026/01/26/neuroinclusive-teaching-in-higher-education/> (Accessed: 3 September 2026).*

### **For Dyslexia**

*University of Strathclyde (n.d.) Barriers faced by students with dyslexia. Available at: <https://www.strath.ac.uk/professionalservices/disabilityandwellbeing/informationforstaff/workingwithstudentswithdyslexia/barriersfacedbystudentswithdyslexia/> (Accessed: 2 July 2026).*

### **For Autism**

*Reachout ASC (n.d.) Autism and behaviour in secondary school. Available at:*

*<https://reachoutasc.com/autism-and-behaviour-in-secondary-school/>*

*(Accessed: 2 July 2026).*

### **For processing disorders**

*National Center for Learning Disabilities (n.d.) Visual and auditory processing disorders.*

*Available at: <https://www.ldonline.org/ld-topics/processing-deficits/visual-and-auditory-processing-disorders> (Accessed: 10 September 2026)*

### **For Dyspraxia**

*Dyspraxia Foundation (2012) Dyspraxia in further and higher education. Available at:*

*<https://dyspraxiafoundation.co.uk/wp-content/uploads/2014/12/Dyspraxia-in-Further-and-Higher-Education-revised-2012.pdf> (Accessed: 10 September 2026).*

### **For ADHD**

*Royal Borough of Windsor and Maidenhead (n.d.) Barriers to learning: poor attention and concentration and/or hyperactivity (ADHD). AfC Info. Available at:*

*[https://rbwm.afcinfo.org.uk/pages/local-offer/information-and-](https://rbwm.afcinfo.org.uk/pages/local-offer/information-and-advice/education/about-sen-support-in-schools/ordinarily-available-provision/barriers-)*

*[advice/education/about-sen-support-in-schools/ordinarily-available-provision/barriers-](https://rbwm.afcinfo.org.uk/pages/local-offer/information-and-advice/education/about-sen-support-in-schools/ordinarily-available-provision/barriers-)*

*to-learning/barrier-to-learning-poor-attention-and-concentration-and-or-hyperactivity-adhd* (Accessed: 10 September 2026).

## **Cardiff University Support services**

<https://www.cardiff.ac.uk/study/student-life/student-support/disability-and-dyslexia-service>